Identifying spatial domains from Spatial Transcriptomics Using Graph Contrastive Learning with an Adversarial Mechanism

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A. The contrastive loss

The contrastive loss is formulated as follows:

$$L_C = \frac{1}{N}\sum_{i=1}^{N} -(log\frac{exp^{sim(H_i^1,Z_i^2)}}{\sum_{k=1,k\neq i}^{N} exp^{sim(H_i^1,Z_k^2)}} + log\frac{exp^{sim(H_i^2,Z_i^1)}}{\sum_{k=1,k\neq i}^{N} exp^{sim(H_i^2,Z_k^1)}})$$

where H_i^1 and H_i^2 define the embeddings of the i -th node in $\boldsymbol{H}_1$ and $\boldsymbol{H}_2$, respectively. Additionally, $sim(\bullet)$ represents the cosine similarity function.

B. Details of dataset

In this study, we utilize five datasets from three different platforms to comprehensively evaluate the performance of STAGCL. Specifically, three datasets are obtained from the 10x Visium platform, including the human dorsolateral prefrontal cortex (DLPFC), human breast cancer (HBC), and the mouse brain anterior (MBA). The DLPFC dataset, manually annotated by Maynard et al. [1], consists of 12 tissue sections, each with 4–6 cortical layers and one ventricle (VM) layer. The HBC dataset is divided into 20 regions by Xu et al. [2]. The MBA dataset is annotated with 52 spatial domains based on the Allen Brain Reference Atlas, presenting a more complex tissue microenvironment compared to the DLPFC and HBC datasets. Additionally, we obtain the mouse somatosensory cortex (MSC) dataset from the osmFISH platform, which is primarily divided into 11 clusters. Finally, the mouse embryo at the 9.5 developmental stage (ME9.5) dataset is derived from the Stereo-seq platform and is partitioned into 12 regions as defined by Chen et al. [3]. Further details of all datasets are summarized in Table 1.

TABLE 1: The statistics of the datasets used in this study.

Datasets	Slices	Spots	Genes	Domains	Platforms
DLPFC	12	3,460-4,789	33,538	5-7	10x Visium
MBA	1	2,695	32,285	52	10x Visium
HBC	1	3798	36,601	20	10x Visium
MSC	1	5328	33	11	osmFISH
ME9.5	1	5,913	22,385	12	Stereo-seq

C. Baseline methods

We select the following state-of-the-art methods:

(1) SCANPY: a powerful and extensible toolkit for analyzing scRNA-seq data.

(2) DeepST [4]: Employs a denoising autoencoder and domain adversarial neural networks (DAN) to capture spot information.

(3) CCST: Utilizes a graph contrastive learning architecture based on Deep Graph Infomax (DGI) [5] for spatial clustering.

(4) STAGATE: Integrates gene expression and spatial topology using a graph attention autoencoder.

(5) GraphST: Leverages a graph self-supervised contrastive learning framework to learn informative latent embeddings.

(6) SEDR: Applies a masked augmentation strategy and uses a variational graph autoencoder [6] to learn representations of gene expression and spatial information.

(7) stDCL [7]: Proposes dual graph contrastive learning to train the model, learning information from expression profiles and spatial contexts.

Among these, SCANPY is a non-spatial clustering method, and the rest are spatial clustering approaches. All baseline methods are implemented using the default parameters as specified in their original paper.

D. Evaluation measurements

To evaluate the effectiveness of STAGCL in spatial domain identification, we adopt several commonly used clustering evaluation metrics. Adjusted rand index (ARI) [8] measures the similarity between predicted labels and ground truth. Normalized mutual information (NMI) [9] quantifies the mutual information between the clustering results and true labels. Homogeneity score (HS) [10] assesses the purity within clusters. Additionally, we apply UMAP [11] to visualize the spatial domains derived from the embeddings learned by STAGCL and baselines.

Spatial domain identification serves as a fundamental step in the downstream analysis of ST data. To further describe the impact of the learned embeddings on downstream tasks, we employ the PAGA [12] algorithm to infer inter-layer developmental trajectories. Moreover, we visualize the gene expression patterns of the embeddings after STGACL denoising and perform differential gene expression analysis (DEGs).

E. Experimental setup

In the data preprocessing stage, we do not apply PCA to the MSC dataset, as each spot contains only 33 genes. During the data augmentation stage, we initialize the reversible network dimension to 200, the mask rate to 0.8, and the edge dropout rate to 0.1. We iterate 600 epochs using the Adam optimizer, with the learning rate and weight decay of 0.0001. During training, the reversible network dimension is 64, and the encoder and decoder dimensions are set to [200, 128, 64] and [64, 128, 200], respectively. The weights for the corresponding loss components— L_C , L_{adv} , L_{rec} , L_M , and L_S —are set to 0.5, 0.1, 1, 1, and 1, respectively. All experiments are conducted on an NVIDIA GeForce RTX 4080 GPU and implemented using PyTorch for the proposed STAGCL model.

F. Ablation studies

Effectiveness of the proposed module

In this section, we remove key components of STAGCL to evaluate their individual contributions. The variants are as follows: STAGCL-w/o-C (removing the contrastive loss L_C); STAGCL-w/o-A (disabling the adversarial loss L_{adv}); STAGCL-w/o-R (omitting the reconstruction loss L_{rec} without the feature reconstruction module) STAGCL-w/o-M (excluding the matching loss L_M without the label matching

mechanism); STAGCL-w/o-S (missing the similarity loss L_S).

The results in Table S1 indicate that STAGCL consistently outperforms its variants, suggesting that each module contributes to the performance improvement. It is worth noting that the contrastive loss exerts a significant impact on the model, particularly for the MBA and MSC datasets, with performance improvements of approximately 10%. This may be attributed to the disruption-and-recovery strategy, which encourages the model to pay more attention to reliable gene expression. Importantly, the contribution of the adversarial mechanism should not be overlooked, as it ensures the diversity of augmented views while maintaining consistency in the embedded latent space. In Fig. S3, we visualize the clustering results of STAGCL and its variants across all 12 slices on the DLPFC dataset, further confirming the importance of each module in improving clustering accuracy.

TABLE S1: Ablation study on the impact of different modules on experiments

	DLPFC		HBC		MBA		ME9.5		MSC	
Method	ARI（mean）	NMI(mean)	ARI	NMI	ARI	NMI	ARI	NMI	ARI	NMI
w/o C	0.5800	0.6663	0.5857	0.6702	0.4383	0.7072	0.2305	0.4666	0.6585	0.7233
w/o A	0.6128	0.6702	0.5946	0.6660	0.4634	0.7236	0.2517	0.4820	0.6880	0.7590
w/o R	0.6195	0.6687	0.5980	0.6612	0.4679	0.6973	0.2548	0.4872	0.6887	0.7578
w/o M	0.6219	0.6718	0.6060	0.6611	0.4913	0.7185	0.2714	0.5049	0.7161	0.7321
w/o S	0.6244	0.6728	0.6123	0.6647	0.5233	0.7123	0.3028	0.5120	0.7450	0.7534
STAGCL	**0.6411**	**0.6751**	**0.6320**	**0.6878**	**0.5305**	**0.7218**	**0.3151**	**0.5163**	**0.7484**	**0.7646**

Ablation on encoder and decoder type

In Table S2, we compare various types of encoder and decoder architectures. The results reveal that the GraphSAGE architecture does not perform well on these datasets, possibly due to its sample construction strategy causes the model to overlook more informative spots. Specifically, we observe that using a simple GCN architecture can significantly improve the performance of STAGCL, further proving the scalability of the method.

TABLE S2: Ablation study on the impact of different graph convolution blocks

GNN		DLPFC（mean）		HBC		MBA		ME9.5		MSC	
Encode	Decode	ARI	NMI	ARI	NMI	ARI	NMI	ARI	NMI	ARI	NMI
GCN	GCN	**0.6411**	**0.6751**	**0.6320**	**0.6878**	**0.5305**	**0.7218**	**0.3151**	**0.5163**	**0.7484**	**0.7646**
GAT	GAT	0.6080	0.6743	0.5823	0.6669	0.5011	0.7122	0.2753	0.5023	0.7104	0.7333
SAGE	SAGE	0.5073	0.5778	0.5969	0.6547	0.4513	0.6942	0.2911	0.5079	0.6828	0.7198
GIN	GIN	0.6166	0.6845	0.6193	0.6978	0.4906	0.7070	0.2554	0.4760	0.7267	0.7563

G. Visualization

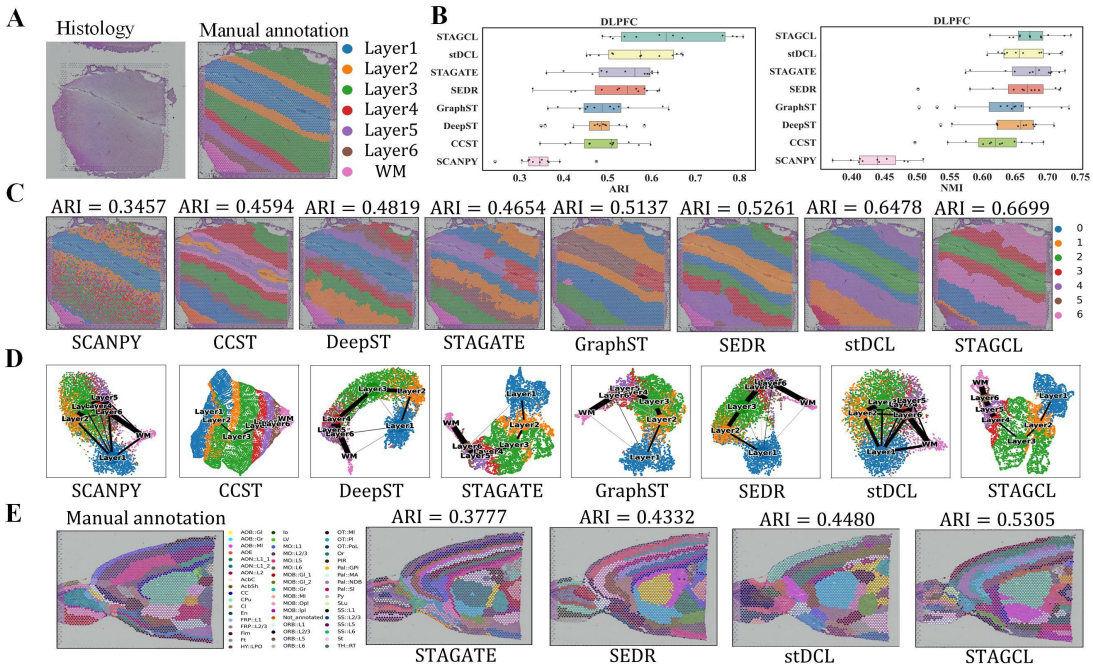

Fig. S1 (A) H&E image and manual annotation in slice 151510. (B) ARI and NMI boxplots of methods across 12 DLPFC slices. (C) The clustering results of STAGCL and baselines for slice 151510. (D) The UMAP and PAGA visualization of the experimental results. (E) Manual annotation (left) and spatial domain identification (right) on MBA. (F) The violin plots of raw expression and denoised expression by STAGCL of marker genes.

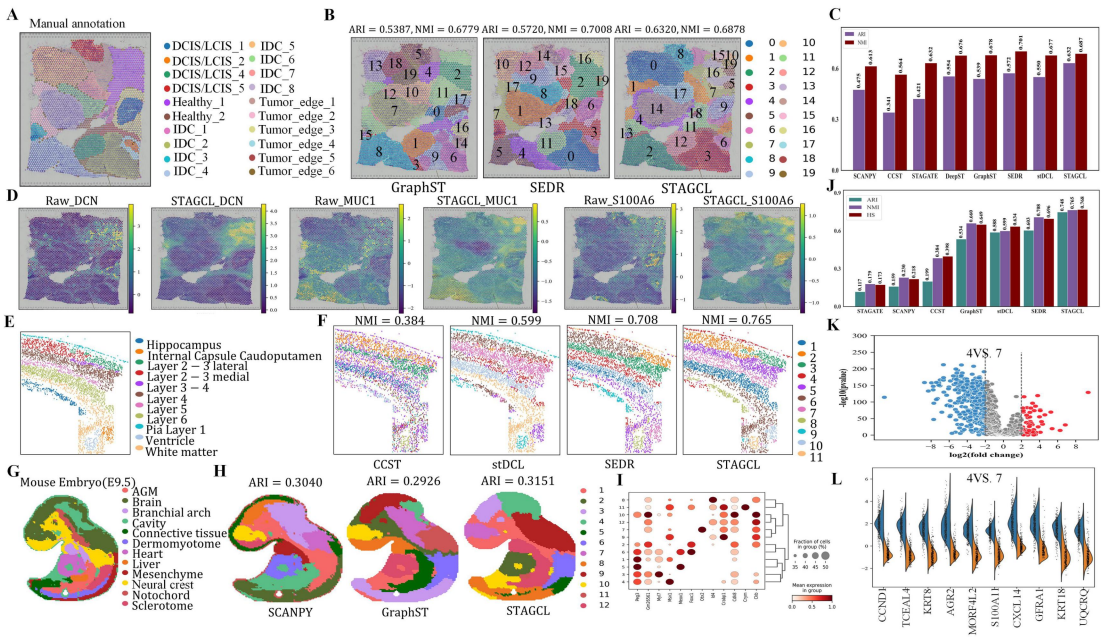

Fig. S2 (A) Manual annotation of the HBC dataset. (B) Spatial domain identification by STAGCL and comparison methods. (C) Bar chart of ARI and NMI metrics for 8 methods. (D) Expression of marker genes on HBC dataset before and after denoising. (E and F) Manual annotation and spatial domains of the MSC dataset. (G and H) Manual annotation of ME9.5 dataset and spatial clusters of various methods. (I) Bubble plots of DEGs revealed by STAGCL. (J) The performance comparison of the STAGCL and 6 baselines. (K and L) Volcano and violin plots of DEGs between cluster pairs (4,7) revealed by STAGCL.

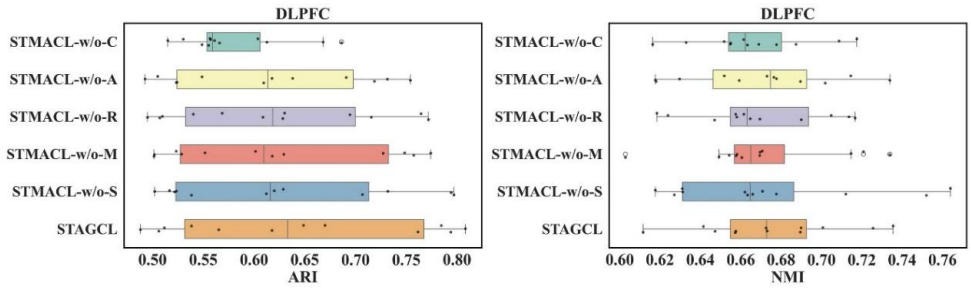

Fig. S3 The ARI and NMI boxplots of STAGCL and its variants on the DLPFC dataset.

H. References

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